triPOS Cloud Retail Transaction Script

triPOS Cloud Communication

triPOS requests should be submitted via the triPOS API to the HTTPS triPOS Cloud URL. triPOS requests requiring processor approval they are then submitted to the Element Express platform for approval. Note that triPOS Cloud requests must be submitted using JSON Format.

triPOS Cloud URL's (Cert Environment)

triPOS Cloud API Documentation: https://triposcert.vantiv.com/api/swagger-ui-bootstrap/
triPOS Cloud Lane Management API Documentation: https://triposcert.vantiv.com/cloudapi/swagger/ui/index
triPOS Cloud API Base URL: https://triposcert.vantiv.com
Express Terminal Settings:
https://developerengine.fisglobal.com/assets/pdf/Express%20Required%20Terminal%20Settings%2009022021.pdf
Supplemental Guide:
https://developerengine.fisglobal.com/assets/pdf/Express%20Supplemental%20Guidelines%20a10012019.pdf
Receipts: https://developerengine.fisglobal.com/apis/tripos/tripos-cloud#receipts

To determine success or failure of the triPOS request, first review the HTTP status code returned. If successful, then review the triPOS statusCode value returned.

Please list the hardware used for this certification:
Hardware connectivity: i.e.: WIFI,IP

Connectivity

triPOS Cloud should occur automatically however to you may wish to confirm within the network configuration that outbound https traffic can reach the following:

- https://tripos.worldpay.com on port **443**
- https://device.worldpay.com on port **9001**
- Enable TCP/IP **TLS v1.2** protocol for https://device.tripos.vantiv.com

tp-authorization header Example: The value is very simple: "Version=1.0"

Two Steps to Processing

1. Pair the device to the merchant's API credentials - When the PIN pad device arrives it will be in an un-paired state. When the device is powered up it will display a seven-character code. Implement a UI element in the business management software to allow a user to enter this six-character code and then send a message to triPOS Cloud to pair the device.
2. Send transactions - After the device is paired to the merchant's API credentials the business management software can send any valid API command to perform transaction processing or other types of processing supported via the API.

Notifications

Please confirm that once your application submits a request to triPOS, your application will wait for a triPOS response (and not take any other payment action until the response is received). triPOS does not support a programmatic cancellation API call. Currently, the only way to cancel a transaction in progress is to press the Cancel button on the physical PINpad. Note that each PINpad screen can be displayed for up to 60 seconds before the PINpad times out, so you may need to adjust your POS timeout length based on the speed of typical transactions with your implementation.

Please confirm that when you receive a response from triPOS, you are first looking at the HTTP status codes returned (http://en.wikipedia.org/wiki/List_of_HTTP_status_codes). If the response is successful, you can then proceed to review the statusCode values returned by triPOS to determine whether the transaction was approved, declined, error, etc. An example of a .Net REST and HTTP client that could be used is RestSharp.

Please ensure that your POS integration does not submit additional requests to the same triPOS Cloud Lane/PIN pad when that Lane/PIN pad is already in the middle of processing a transaction or interacting with the cardholder. The original triPOS Cloud transaction must be 100% complete before additional requests are submitted. Any additional requests submitted to an in-progress Lane/PIN pad will likely cause triPOS Cloud errors to be returned to the POS application.

NEED TO INCLUDE.

Reference Number

The ReferenceNumber is an input value that is defined by the merchant or software application. The ReferenceNumber can be used to identify the transaction/order and should be included with each transaction request submitted on the Express platform. We recommend this value be a unique, numeric value up to 16 digits.

Ticket Number

The TicketNumber is an input value that is defined by the merchant or software application, and it is used for interchange purposes. Some value should be submitted in this field to help the merchant transaction obtain the lowest interchange rate, and though not required, we recommend this value be a unique, numeric value up to 6 digits. A good example of this is a 6-digit incremental value that can be recycled upon exceeding 999999.

Instructions

Please submit the appropriate Card test requests for the triPOS methods to be supported within your application along with any methods specifically requested by the Element certification team. Also, please let the Certification team know if you plan to support Tip, Cash Back, or Convenience Fee Amounts.

POST /api/v1/sale

Sale https://developerengine.fisglobal.com/apis/tripos/tripos-cloud/tripos-transaction/api-specification#tag/Sale

Submit the appropriate **Sale** test transactions.

Test Cases	Amount	TransactionID	Comments and Feedback
CP Swiped Credit Card	1.04		
triPOS Cloud Request ID Used			
CP Swiped Credit Card (Partial Approval—allowPartialApprovals)	9.65		
triPOS Cloud Request ID Used			
CP Swiped Credit Card (Balance Response)	32.00		
triPOS Cloud Request ID Used			
CP Swiped PIN Debit Card	31.00		
triPOS Cloud Request ID Used			
CP Swiped PIN Debit Card (with 1.00 Cash Back)	31.00		
triPOS Cloud Request ID Used			
CP EMV Card Insert	1.06		
triPOS Cloud Request ID Used			
CP Contactless EMV Credit Card	1.08		
triPOS Cloud Request ID Used			
CP Keyed Credit Card	1.07		
triPOS Cloud Request ID Used			
CP Keyed Credit Card (Partial Approval—allowPartialApprovals)	9.65		
triPOS Cloud Request ID Used			
CP Keyed Credit Card (Balance Response)	32.00		
triPOS Cloud Request ID Used			

POST /api/v1/authorization

https://developerengine.fisglobal.com/apis/tripos/tripos-cloud/tripos-transaction/api-specification#tag/Authorization

Submit the appropriate **Authorization** test transactions.

Test Cases	Amount	TransactionID	Comments and Feedback
CP Swiped Credit Card	1.04		
triPOS Cloud Request ID Used			
CP Swiped Credit Card (Partial Approval—allowPartialApprovals)	9.65		
triPOS Cloud Request ID Used			
CP Swiped Credit Card (Balance Response)	32.00		
triPOS Cloud Request ID Used			
CP Contactless EMV Credit Card	1.08		
triPOS Cloud Request ID Used			
CP EMV Card Insert	1.06		
triPOS Cloud Request ID Used			
CP Keyed Credit Card	1.07		
triPOS Cloud Request ID Used			
CP Keyed Credit Card (Partial Approval—allowPartialApprovals)	9.65		
triPOS Cloud Request ID Used			

CP Keyed Credit Card (Balance Response)	32.00		
triPOS Cloud Request ID Used			
Notes (Element Use Only)			

POST /api/v1/authorization/{transactionId}/completion

https://developerengine.fisglobal.com/apis/tripos/tripos-cloud/tripos-transaction/api-specification#tag/Authorization/operation/postv1authorizationtransactionIdcompletion

Submit the appropriate **Authorization Completion** test transactions.

Test Cases	Amount	TransactionID	Comments and Feedback
Completion of Swiped Credit Card	1.04		
triPOS Cloud Request ID Used			
Completion of Swiped Credit Card (Partial Approval)	6.10		
triPOS Cloud Request ID Used			
Completion of Swiped Credit Card (Balance Response)	32.00		
triPOS Cloud Request ID Used			
Completion of Contactless EMV Credit Card	1.08		
triPOS Cloud Request ID Used			
Completion of EMV Card Insert	1.06		
triPOS Cloud Request ID Used			
Completion of Keyed Credit Card	1.07		
triPOS Cloud Request ID Used			
Completion of Keyed Credit Card (Partial Approval)	6.10		
triPOS Cloud Request ID Used			
Keyed Credit Card (Balance Response)	32.00		
triPOS Cloud Request ID Used			

POST /api/v1/refund

https://developerengine.fisglobal.com/apis/tripos/tripos-cloud/tripos-transaction/api-specification#tag/Refund

Submit the appropriate **Refund** test transactions.

Test Cases	Amount	TransactionID	Comments and Feedback
CP Swiped Credit Card	1.12		
triPOS Cloud Request ID Used			
CP Swiped PIN Debit Card	31.00		
triPOS Cloud Request ID Used			
CP Contactless EMV Credit Card	2.31		
triPOS Cloud Request ID Used			
CP EMV Card Insert	2.32		
triPOS Cloud Request ID Used			
CP Keyed Credit Card	1.13		
triPOS Cloud Request ID Used			
Notes (Element Use Only)			

POST /api/v1/return/{transactionId}/{paymentType}

https://developerengine.fisglobal.com/apis/tripos/tripos-cloud/tripos-transaction/api-specification#tag/Return

Submit the appropriate **return** (refund) test transactions.

Test Cases	Amount	TransactionID	Comments and Feedback
Return of Swiped Credit Card	1.04		
triPOS Cloud Request ID Used			
Return of Contactless EMV Credit Card	1.08		
triPOS Cloud Request ID Used			
Return of Keyed Credit Card	1.07		
triPOS Cloud Request ID Used			
Partial Return of Swiped Credit Card	0.50		
triPOS Cloud Request ID Used			
Partial Return of Keyed Credit Card	0.53		
triPOS Cloud Request ID Used			
Notes (Element Use Only)			

POST /api/v1/reversal/{transactionId}/{paymentType}

https://developerengine.fisglobal.com/apis/tripos/tripos-cloud/tripos-transaction/api-specification#tag/Reversal

Submit the appropriate **Reversal** test transactions (full reversal).

Test Cases	Amount	TransactionID	Comments and Feedback
Full Reversal of Swiped Credit Card	1.00		
triPOS Cloud Request ID Used			
Full Reversal of Swiped PIN Debit Card	31.00		
triPOS Cloud Request ID Used			
Full Reversal of Contactless MSD Credit Card	1.00		
triPOS Cloud Request ID Used			
Full Reversal of EMV Card Insert	1.00		
triPOS Cloud Request ID Used			
Full Reversal of Keyed Credit Card	1.00		
triPOS Cloud Request ID Used			
Full Reversal via Direct Express (If original Sale was submitted via triPOS, and if you then wish to Reverse the transaction directly on Express, please verify that you submit the same LaneNumber on your Express Reversal request)	any amount		
Notes (Element Use Only)			

POST /api/v1/void/{transactionId}

https://developerengine.fisglobal.com/apis/tripos/tripos-cloud/tripos-transaction/api-specification#tag/Void

Submit the appropriate **Void** test transactions.

Test Cases	Amount	TransactionID	Comments and Feedback
Void of Swiped Credit Card	1.00		
triPOS Cloud Request ID Used			
Void of Contactless EMV Credit Card	1.00		
triPOS Cloud Request ID Used			
Void of EMV Card Insert	1.00		
triPOS Cloud Request ID Used			
Void of Keyed Credit Card	1.00		
triPOS Cloud Request ID Used			
Notes (Element Use Only)			

POST /api/v1/force/credit

https://developerengine.fisglobal.com/apis/tripos/tripos-cloud/tripos-transaction/api-specification#tag/Force

Submit the appropriate **Force** test transactions (this is a credit card transaction type).

Test Cases	Amount	TransactionID	Comments and Feedback
CP Swiped Credit Card	3.10		
triPOS Cloud Request ID Used			
CP Contactless EMV Credit Card	2.31		
triPOS Cloud Request ID Used			
CP Keyed Credit Card	3.13		
triPOS Cloud Request ID Used			
Notes (Element Use Only)			

GET /api/v1/binQuery/{laneId}

https://developerengine.fisglobal.com/apis/tripos/tripos-cloud/tripos-transaction/api-specification#tag/BinQuery

Submit the appropriate **BinQuery** test transactions.

Test Cases	Amount	TransactionID	Comments and Feedback
Swiped Credit Card	N/A		N/A
triPOS Cloud Request ID Used			
Contactless EMV Credit Card	N/A		N/A
triPOS Cloud Request ID Used			
Keyed Credit Card	N/A		N/A
triPOS Cloud Request ID Used			
Notes (Element Use Only)			

Level 2 Processing

https://developerengine.fisglobal.com/apis/express/features/level-2-data#express-level-ii-data

To help qualify for the lowest interchange rates on commercial cards, we recommend that Level II
transactions include the following:

- SalesTaxAmount (valid SalesTaxAmount must be greater than 0.1% of the sale amount, as taxexempt transactionsdo not qualify for Level II interchange rates)
- CommercialCardCustomerCode (such as a PO Number)
- ShippingZipcode
- TicketNumber
- BillingName

POST /api/v1/sale with Level 2 information

Test Cases	Amount	TransactionID	Comments and Feedback
CP Swiped Credit Card	3.00		
triPOS Cloud Request ID Used			
CP EMV Card Insert	4.00		
triPOS Cloud Request ID Used			

POST /api/v1/authorization with Level 2 information

Test Cases	Amount	TransactionID	Comments and Feedback
CP Swiped Credit Card	5.00		
triPOS Cloud Request ID Used			
CP EMV Card Insert	6.00		
triPOS Cloud Request ID Used			

Duplicate Transaction Handling

Submit the appropriate test transactions for handling duplicate transactions.

Duplicate Checking

Duplicate Checking is a host-based feature that can be turned on or off per transaction. It provides a means to identify duplicate transactions by the following parameters:

- Same Card Number
- Same Transaction Amount
- Specific Transaction Category (Sale, Return, etc.)
- Within a certain number of Transactions (25) in the same batch

There are two methods that can be used to either override or disable duplicate checking:

- DuplicateOverrideFlag: Set this flag to true to override the duplicate checking feature if the response from the host indicates a duplicate transaction has been attempted. In other words, if you send a transaction that was determined to be a duplicate by the host, set this flag to true and resend the same transaction to bypass the duplicate checking feature on the host. This is a per transaction request feature.
- DuplicateCheckDisableFlag: Set this flag to true to disable the duplicate checking feature on the host. If this flag is set to true no duplicate checks will be performed for that transaction. This is a per transaction request feature.

Since all per-transaction duplicate checking enable / disable functionality can be accomplished with the single DuplicateCheckDisableFlag parameter, it is recommended that integrators move forward with implementing only the DuplicateCheckDisableFlag parameter.

SAME CARD FOR ALL TRANSACTIONS BELOW

Test Cases	Amount	TransactionID	Comments and Feedback
Process Sale transaction or Auth	1.70		
Process duplicate Sale transaction (checkForDuplicateTransactions) if you receive a 23 response. Do test case below.	1.70		
triPOS Cloud Request ID Used			
CP Swiped Credit Card or EMV with either DuplicateOverrideFlag	1.70		

or DuplicateCheckDisableFlag as True			

Mandatory End Points for Lane Management

Lane Management	triPOS Request ID
POST /cloudapi/v1/lanes/	
DELETE /cloudapi/v1/lanes/{laneId}	

Other triPOS Endpoints

Please identify other triPOS endpoints you intend to utilize.

Other triPOS Cloud Endpoints	triPOS Request ID
POST /api/v1/display	
POST /api/v1/idle	
GET /api/v1/input/{laneId}	
GET /api/v1/selection/{laneId}	
GET /api/v1/signature/{laneId}	
GET /api/v1/status/host	
GET /api/v1/status/triPOS/{echo}	
GET /cloudapi/v1/lanes/{laneId}/connectionstatus	
When using supported triPOS Cloud endpoints to perform a one of the status check requests defined, please note that whichever requests are chosen should be submitted at a maximum rate of no more than one (1) status check every two (2) minutes. This requirement is in place to ensure stability of the triPOS Cloud platform. Status check submissions at higher rates may force Worldpay/FIS to restrict (reject) merchant processing for periods of time.	